

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester 2)
Scala For DS

Practical - 30

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Class : SYBSC IT	Batch : 1st
Date of Assignment : 29/08/2026	Date/Time of Submission : 29/08/2026

Practical 30

Aim : Cluster a dataset into two groups using k-means clustering in Breeze.

Code :

```
build.sbt
```

```
libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
```

```
import breeze.linalg._
```

```
object Demo extends App {
```

```
  // Dataset
```

```
  val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)
```

```
  // Initial cluster centers
```

```
  var center1 = 2.0
```

```
  var center2 = 10.0
```

```
  // Repeat clustering
```

```
  for (i <- 1 to 5) {
```

```
    val cluster1 = data.toArray.filter(x =>
```

```
      math.abs(x - center1) < math.abs(x - center2)
```

```
    )
```

```
    val cluster2 = data.toArray.filter(x =>
```

```

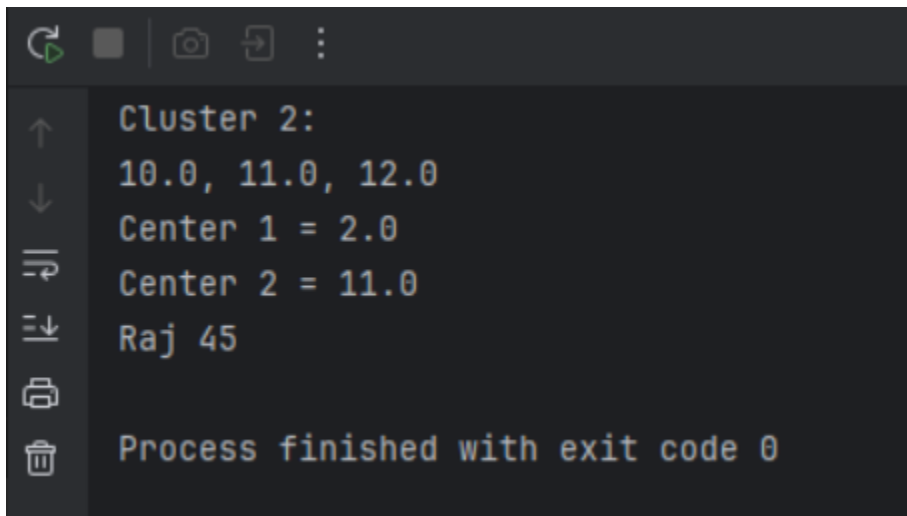
    math.abs(x - center2) <= math.abs(x - center1)
  )
  // Calculate new centers
  center1 = cluster1.sum / cluster1.length
  center2 = cluster2.sum / cluster2.length
}
// Display clusters
println("Cluster 1:")
println(data.toArray.filter(x =>
  math.abs(x - center1) < math.abs(x - center2)
).mkString(", "))

println("Cluster 2:")
println(data.toArray.filter(x =>
  math.abs(x - center2) <= math.abs(x - center1)
).mkString(", "))

println("Center 1 = " + center1)
println("Center 2 = " + center2)
}

```

Output :



```

Cluster 2:
10.0, 11.0, 12.0
Center 1 = 2.0
Center 2 = 11.0
Raj 45

Process finished with exit code 0

```

